

AI Automation Internship – Practical Assignment 1 Prompt Engineering

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Program: BSCS

University: Federal Urdu University of Arts, Science & Technology , Islamabad

Internship: AI Automation Internship

Q1 : Learn how to write effective prompts and understand how prompt structure affects AI output.

Task

Choose **one topic** such as:

- Education
- Business
- Healthcare
- Technology
- Social Media
- Career Guidance

Create prompts using the following prompt-engineering concepts:

1. **Clarity**
2. **Context**
3. **Role**
4. **Output Format**
5. **Tone**
6. **Constraints**

Introduction

Prompt Engineering is the process of creating effective instructions for AI systems to get better, more accurate, and more useful results. In this assignment, I selected Technology as my topic and experimented with different prompting techniques.

Part A – Basic vs Improved Prompt

1. Original / Basic Prompt

First, I created a simple prompt without giving the AI many instructions.

My Prompt

Tell me about Artificial Intelligence and how it is used in technology.

AI Output

Artificial Intelligence (AI) is a technology that allows computers and machines to perform tasks that normally require human intelligence.

AI is used in many areas of technology, including chatbots, recommendation systems, cybersecurity, healthcare, robotics, self-driving cars, and software development.

For example, AI chatbots can understand questions and provide answers, while recommendation systems can suggest movies, products, or videos based on user interests.

AI is becoming an important technology because it can automate tasks, analyze large amounts of data, recognize patterns, and help people solve problems.

My Observation

The answer was correct and useful, but it was quite general. I did not tell the AI who I was, what level of explanation I needed, how I wanted the answer structured, or how long the answer should be.

2. Improved / Professional Prompt

After testing the basic prompt, I improved it by using the main prompt-engineering concepts mentioned in the assignment.

My Improved Prompt

Role: You are an experienced AI and Technology teacher.

Context: I am Sayyam Shahid, a BSCS student learning Artificial Intelligence and AI Automation. I want to understand the role of Artificial Intelligence in modern technology in a simple and practical way.

Task: Explain what Artificial Intelligence is and how it is used in modern technology. Explain its important applications and provide real-world examples.

Output Format: Structure the answer using the following sections:

1. What is Artificial Intelligence?
2. How does AI work?
3. Applications of AI in Technology
4. Real-world examples
5. Benefits of AI
6. Future of AI in Technology

Tone: Use a professional, educational, and beginner-friendly tone.

Constraints:

1. **Use simple English.**
2. **Avoid unnecessary technical terms.**
3. **Explain technical terms when necessary.**
4. **Use headings and bullet points.**
5. **Keep the answer between 500 and 700 words.**
6. **Use practical examples.**

AI Output

1. What is Artificial Intelligence?

Artificial Intelligence (AI) is a field of computer science that focuses on creating computer systems that can perform tasks that normally require human intelligence.

These tasks include learning, understanding language, recognizing images, identifying patterns, making predictions, and solving problems.

For example, an AI chatbot can understand a user's question and generate a relevant response.

2. How Does AI Work?

AI systems use algorithms, data, and computing power to learn patterns and produce useful results.

Machine Learning is an important part of AI. In Machine Learning, a computer model learns from existing data and uses what it has learned to make predictions or decisions about new data.

3. Applications of AI in Technology

AI Chatbots : AI chatbots can communicate with users using natural language.

Example: Customer-support chatbots can answer common questions automatically.

Recommendation Systems :AI can analyze user behavior and recommend relevant content.

Example: A streaming platform can recommend movies based on a user's previous activity.

Computer Vision : Computer Vision allows computers to analyze images and videos.

Example: AI can be used to detect objects in photographs or security-camera footage.

Cybersecurity : AI can analyze network activity and identify unusual patterns.

Example: AI systems can help detect suspicious activities in computer networks.

Software Development : AI tools can assist developers with writing, debugging, testing, and documenting code.

Example: An AI coding assistant can generate a function based on a developer's instructions.

4. Benefits of AI : Some important benefits of AI include:

- Automation of repetitive tasks
- Faster data analysis
- Pattern recognition
- Personalized user experiences
- Support for decision-making
- Increased productivity

5. Future of AI in Technology

AI is becoming an important part of modern technology. It is being used in software development, education, healthcare, business, cybersecurity, robotics, and many other fields.

In the future, AI and automation can work together to create more intelligent software applications, AI agents, and automated systems.

My Observation

The improved prompt gave me a much more structured and relevant answer. Because I provided the AI with my background as a BSCS student, it was able to give an explanation suitable for my learning level

Prompt Engineering Concepts I Used :

Concept	How I Used It
Clarity	I clearly explained what I wanted the AI to discuss.
Context	I explained that I am a BSCS student learning AI and AI Automation.
Role	I asked the AI to act as an experienced AI and Technology teacher.
Output Format	I specified the exact sections and structure for the answer.
Tone	I requested a professional , educational, and beginner-friendly tone.
Constraints	I specified simple English, word count, headings, and practical examples.

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Part B – Three Prompt Techniques :

After testing the basic prompt, I improved it by using the main prompt-engineering concepts mentioned in the assignment.

My Improved Prompt

Role:

You are an experienced AI and Technology teacher.

Context:

I am Sayyam Shahid, a BSCS student learning Artificial Intelligence and AI Automation. I want to understand the role of Artificial Intelligence in modern technology in a simple and practical way.

Task:

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Tone:

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6. Use practical examples.

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These tasks include learning, understanding language, recognizing images, identifying patterns, making predictions, and solving problems.

For example, an AI chatbot can understand a user's question and generate a relevant response.

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Machine Learning is an important part of AI. In Machine Learning, a computer model learns from existing data and uses what it has learned to make predictions or decisions about new data.

3. Applications of AI in Technology

AI Chatbots

AI chatbots can communicate with users using natural language.

Example: Customer-support chatbots can answer common questions automatically.

Recommendation Systems

AI can analyze user behavior and recommend relevant content.

Example: A streaming platform can recommend movies based on a user's previous activity.

Computer Vision

Computer Vision allows computers to analyze images and videos.

Example: AI can be used to detect objects in photographs or security-camera footage.

Cybersecurity

AI can analyze network activity and identify unusual patterns.

Example: AI systems can help detect suspicious activities in computer networks.

Software Development

AI tools can assist developers with writing, debugging, testing, and documenting code.

Example: An AI coding assistant can generate a function based on a developer's instructions.

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Some important benefits of AI include:

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In the future, AI and automation can work together to create more intelligent software applications, AI agents, and automated systems.

My Observation

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Part B – Three Prompt Techniques

1. Zero-Shot Prompting

Definition

Zero-shot prompting means asking the AI to perform a task **without providing any examples**.

My Prompt

Explain five important applications of Artificial Intelligence in modern technology.

For each application, provide:

- Application name
- Short explanation
- One real-world example

Use simple English and organize the answer using headings and bullet points.

AI Output

1. Chatbots

Explanation: AI chatbots communicate with users and answer questions using natural language.

Example: Customer-support chatbots can answer common customer questions.

2. Recommendation Systems

Explanation: AI analyzes user behavior to recommend relevant content.

Example: Streaming platforms recommend movies and shows based on viewing history.

3. Computer Vision

Explanation: Computer Vision allows computers to understand information from images and videos.

Example: AI can detect objects in security-camera footage.

4. Cybersecurity

Explanation: AI can identify unusual patterns that may indicate security threats.

Example: Security systems can use AI to detect suspicious network activity.

5. Software Development

Explanation: AI can assist developers in coding, debugging, testing, and documentation.

Example: AI coding assistants can generate code from natural-language instructions.

My Observation

The AI understood the task without receiving any examples. This made the prompt simple and direct.

2. Few-Shot Prompting

Definition

Few-shot prompting means providing the AI with **a few examples** before asking it to perform the actual task.

My Prompt

Classify each statement as either "AI Technology" or "Non-AI Technology".

Example 1:

Statement: A chatbot understands questions and provides answers.

Answer: AI Technology

Example 2:

Statement: A basic calculator calculates $10 + 20$.

Answer: Non-AI Technology

Example 3:

Statement: A recommendation system suggests products based on customer behavior.

Answer: AI Technology

Now classify the following:

1. A system detects faces in photographs.
2. A digital clock displays the current time.
3. A system predicts whether an email is spam.
4. A text editor allows users to change font size.

Output only the number and classification.

AI Output

1. AI Technology
2. Non-AI Technology
3. AI Technology
4. Non-AI Technology

My Observation

The examples helped the AI understand exactly how I wanted the statements to be classified and how the final answer should be formatted.

3. Step-by-Step / Structured Reasoning

Definition

Structured reasoning means dividing a complex task into clear stages or sections so the AI can produce a systematic answer.

My Prompt

You are an AI Technology consultant.

A university wants to develop an AI chatbot for students.

The chatbot should answer questions about courses, class schedules, assignments, university services, and frequently asked questions.

Analyze this project using the following structure:

1. Problem
2. Proposed AI Solution
3. Required Data
4. Main Features
5. Benefits
6. Limitations
7. Security and Privacy
8. Conclusion

AI Output

1. Problem

- Students may have difficulty finding university information.
- Staff may have to answer the same questions repeatedly.
- Information may be available in different places.

2. Proposed AI Solution

- Develop an AI-powered chatbot.
- Allow students to ask questions using natural language.
- Connect the chatbot to approved university information.

3. Required Data

- Course information
- Academic schedules
- Frequently asked questions

- University service information
- Approved university policies

4. Main Features

- Natural-language questions and answers
- Course information
- FAQ support
- 24/7 availability
- Human support escalation

5. Benefits

- Faster access to information
- Reduced repetitive work
- Better student support
- Automated responses to common questions

6. Limitations

- The chatbot may misunderstand some questions.
- Incorrect data can produce incorrect answers.
- Complex problems may require human assistance.

7. Security and Privacy

- Student information should be protected.
- Sensitive information should not be publicly exposed.
- Access controls should be implemented.
- The chatbot's data should be regularly reviewed.

8. Conclusion

An AI chatbot can help a university provide faster access to common information. It should use reliable and updated information and provide human support when necessary.

My Observation

Structured reasoning was useful for a complex problem because it divided the problem into smaller sections. This made the final answer easier to understand and analyze.

Part C – Testing

Run your prompts in any AI tool such as ChatGPT, Gemini, or Claude.

Compare the results and explain:

- Which prompt produced the best result?
- What changed after improving the prompt?
- Which instructions had the biggest effect?

Submission

Submit:

- Original prompt
 - Improved prompt
 - Three prompt-technique examples
 - AI outputs
 - A short comparison of approximately **200–300 words**
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Gemini :

Role: You are an experienced AI and Technology teacher.

Context: I am Sayyam Shahid, a BSCS student

learning Artificial Intelligence and AI Automation. I want to understand the role of Artificial Intelligence in modern technology in a simple and practical way.

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- 2.How does AI work?
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- 4.Real-world examples
- 5.Benefits of AI
- 6.Future of AI in Technology

Tone: Use a professional, educational, and beginner-friendly tone.

Constraints:

1. Use simple English.
2. Avoid unnecessary technical terms.
3. Explain technical terms when necessary.
4. Use headings and bullet points.
5. Keep the answer between 500 and 700 words.
7. Use practical examples.

Output:

1. What is Artificial Intelligence?

Artificial Intelligence (AI) is a branch of computer science focused on creating smart machines that can perform tasks that usually require human intelligence.

Instead of writing explicit line-by-line instructions for every scenario, we train systems to recognize patterns, make decisions, solve problems, and learn from experience. In modern technology, AI acts as an intelligent engine that makes software dynamic rather than rigid.

2. How does AI work?

At its core, modern AI relies on data and mathematical algorithms. Here is a simplified breakdown of the process:

- **Data Collection:** The system feeds on large datasets, such as images, text, or user click histories.

- **Pattern Recognition (Machine Learning):** The algorithm analyzes this data to find recurring trends or rules.
- **Prediction & Decision:** Based on those patterns, the system makes predictions or completes a task.
- **Feedback & Learning:** When the system receives new data or feedback on its errors, it adjusts its internal rules to get better over time.

3. Applications of AI in Technology

AI drives many of the core software architectures and automated systems used today:

- **Natural Language Processing (NLP):** Enables computers to read, interpret, and generate human language.
- **Computer Vision:** Allows software to process, analyze, and "see" visual inputs like photos and video feeds.
- **Robotic Process Automation (RPA):** Combines AI with software bots to automate repetitive visual or data-entry tasks.
- **Predictive Analytics:** Uses historical data to forecast future outcomes, such as server load or market trends.

4. Real-world examples

Here are practical examples of AI in daily tech products:

- **Generative AI & Assistants:** Systems like ChatGPT and Gemini process natural language prompts to write code, draft articles, and answer questions.
- **Smart Recommendation Engines:** Platforms like YouTube, Netflix, and Spotify analyze your viewing habits to recommend videos or songs tailored to your taste.
- **Self-Driving Vehicles:** Tesla and Waymo use computer vision and sensors to detect pedestrians, traffic signals, and lane markers in real time.
- **Automated Customer Support:** AI chatbots handle standard support tickets instantly, routing complex issues to human agents.

5. Benefits of AI

Adding AI to software systems provides several key advantages:

- **High Efficiency:** AI handles repetitive tasks at scale without getting tired, freeing developers to focus on architecture and strategy.
- **Speed and Accuracy:** Machine learning models can analyze millions of data points in seconds to catch errors or security threats human eyes might miss.
- **24/7 Availability:** Automated AI systems run continuously without downtime.

- **Personalization:** Enables products to adapt uniquely to each user's preferences automatically.

6. Future of AI in Technology

As you move forward in your BSCS degree, you will see AI transition from basic automation to deep integration:

- **Hyper-Automation:** Entire business workflows—from software testing to supply chain management—will run autonomously.
- **AI-Assisted Coding:** Tools like GitHub Copilot will become standard companions for software engineers, speeding up development cycles.
- **Edge AI:** AI models will become light enough to run directly on smaller devices (like phones or IoT sensors) without relying on cloud connections.

Claude Output:

1. What is Artificial Intelligence?

Artificial Intelligence (AI) is the ability of a computer or machine to perform tasks that normally require human intelligence — such as understanding language, recognizing images, making decisions, and learning from experience.

In simple words: **AI is about teaching machines to think and act smartly**, similar to how humans do, but using data and mathematical models instead of a brain.

- AI is not a single technology — it's a broad field.
- It includes smaller areas like **Machine Learning (ML)** (machines learning from data) and **Deep Learning** (a more advanced form of ML inspired by the human brain's neural networks).

2. How does AI work?

AI systems generally work in three main steps:

- **Data Collection** – AI needs large amounts of data (text, images, numbers) to learn patterns. For example, an AI that detects spam emails learns from thousands of example emails.
- **Learning (Training)** – Using algorithms, the AI studies the data and identifies patterns. This is where "Machine Learning" comes in — the machine improves its performance by learning from examples instead of being explicitly programmed for every situation.

- **Prediction/Decision Making** – Once trained, the AI applies what it learned to new, unseen data. For example, it can predict whether a new email is spam or not spam.

Think of it like teaching a child using examples: the more correct examples they see, the better they get at recognizing patterns on their own.

3. Applications of AI in Technology

AI is now part of almost every modern technology field:

- **Automation** – AI-powered bots handle repetitive tasks like data entry, customer support, and workflow automation (relevant to your interest in AI Automation).
- **Natural Language Processing (NLP)** – Helps machines understand and generate human language (used in chatbots, translators, and voice assistants).
- **Computer Vision** – Allows machines to "see" and interpret images or videos (used in face recognition, medical scans).
- **Robotics** – Combines AI with machines to perform physical tasks (used in manufacturing, drones).
- **Recommendation Systems** – Suggests content or products based on user behavior.
- **Cybersecurity** – Detects unusual patterns to identify fraud or hacking attempts.

4. Real-world examples

- **Netflix & YouTube** – Use AI to recommend movies/videos based on your watching history.
- **Google Maps** – Uses AI to predict traffic and suggest the fastest route.
- **ChatGPT / Claude** – Use AI (specifically Large Language Models) to understand and respond to human questions.
- **Self-driving cars (Tesla)** – Use AI with computer vision to detect roads, obstacles, and pedestrians.
- **Banking Apps** – Use AI to detect fraudulent transactions in real time.
- **Face ID (iPhones)** – Uses AI-based facial recognition to unlock phones.
- **AI Automation tools (like Zapier, n8n, or UiPath)** – Automatically move data between apps, send emails, or generate reports without human effort — a great example of AI Automation you're studying.

5. Benefits of AI

- **Saves time** by automating repetitive tasks.
- **Improves accuracy** by reducing human error (e.g., in medical diagnosis).

- **Available 24/7**, unlike humans who need rest.
- **Handles large data** that would take humans years to analyze.
- **Personalizes experiences**, such as recommendations and customer support.
- **Boosts productivity** in businesses by automating workflows.

6. Future of AI in Technology

AI is expected to become even more integrated into daily life and industries:

- **Smarter Automation** – Businesses will rely more on AI agents to handle complete workflows without human input.
- **AI + IoT (Internet of Things)** – Smart homes and smart cities will use AI to manage devices automatically.
- **Healthcare AI** – More accurate disease detection and personalized treatment plans.
- **AI in Education** – Personalized learning experiences based on a student's pace and strengths (something directly relevant to your BSCS journey).
- **Autonomous Systems** – More advanced self-driving vehicles, drones, and robots.
- **Human-AI Collaboration** – Instead of replacing humans, AI will work alongside people to boost efficiency, especially in fields like AI Automation, where routine tasks are handled by AI while humans focus on strategy and creativity.

In short: AI is transforming technology by making systems smarter, faster, and more efficient. As a BSCS student focusing on AI and Automation, understanding these basics will help you build practical projects — like chatbots, automation workflows, or recommendation systems — that solve real-world problems.

Part C – Testing and Comparison

1st Position: Claude

For someone who wants the best output quality regardless of task, one comparison ranked it as: Claude, then ChatGPT, then Gemini. Claude was also noted to deliver the most nuanced reasoning, particularly for complex multi-step problems, and to produce the strongest code with fewer hallucinations.

2nd Position: ChatGPT

Strong across the board — most versatile, best plugin/IDE ecosystem, and the widest general knowledge base. It remains the most widely adopted tool.

3rd Position: Gemini

Called "the quiet all-rounder" — never dominates any single category, but is consistently solid in second or first place, especially for research, multimodal tasks (image/video), and Google Workspace integration.

THE END